

SE LAB 2: Reflection Questions

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Project: Warehouse Inventory & Pallet Location Tracker (WIPLT)

Sprint summary (from the Jira board)

Sprint	Story	Epic	Points	Priority
1	1.1 Scan Pallet ID	Pallet Identification	5	High
1	2.1 Validate Rack Capacity	Pallet Placement & Load Validation	8	High
1	3.1 Log Stock Movements	Stock Movement Tracking & Audit	5	High
1	3.2 Immutable Audit Log	Stock Movement Tracking & Audit	8	High
1	4.2 Fast, Secure Lookup	Location Lookup & Retrieval	8	High
2	2.2 Request Placement Override	Pallet Placement & Load Validation	5	Medium
2	4.1 Lookup Pallet / SKU Location	Location Lookup & Retrieval	3	Medium

Sprint 1 committed 34 story points across 5 stories. Sprint 2 committed 8 story points across 2 stories. Both sprints closed with every committed item in Done.

1. Did your estimations reflect the actual effort?

Partly. The relative sizing held up well. The stories I rated 8 points (Validate Rack Capacity, Immutable Audit Log and Fast, Secure Lookup) are the ones with real complexity behind them: capacity rules, tamper-proof logging and indexed, access-controlled search. The 5-point stories (Scan Pallet ID, Log Stock Movements) are more direct, and Lookup Pallet / SKU Location at 3 points is mostly a read query on top of work already done in Sprint 1.

In the simulation itself, though, the 8-point stories did not take noticeably longer to close than the 5-point ones, so the board cannot confirm the absolute numbers. Since the sprint was run in a compressed time window, the estimates were tested for consistency with each other, not against real hours of work. In a real sprint I would expect the audit log and the secure lookup to be the stories most likely to overrun, because both depend on security and storage decisions that are easy to underestimate.

2. Was your backlog well-prioritized?

Yes. All five Sprint 1 stories were marked High priority and they form the core flow of the system: identify a pallet, check that it can be placed, record the movement, keep an audit trail of it and look it up again. Nothing in Sprint 2 could work without these, so building them first was the right order.

Sprint 2 held the two Medium-priority stories. Request Placement Override is an exception path that only makes sense once normal placement validation exists, and Lookup Pallet / SKU Location extends the lookup work from Story 4.2. Both depend on Sprint 1, which confirms the ordering. One thing I would change is splitting the epics more evenly: Epic 1 (Pallet

Identification) had only one story, while Epic 2 and Epic 4 were spread across both sprints, which made the epic-level progress harder to read.

3. How did your simulated sprint align with your plan?

It aligned closely. Every story that was committed to a sprint was completed in that sprint, with no carry-over and no scope added mid-sprint. Sprint 1 moved items through To Do, In Progress, In Review and Done as planned; the mid-sprint board shows one story done, one in progress, one in review and two still in To Do, which is a normal spread for a healthy sprint. Sprint 2 started only after Sprint 1 was closed and both its stories were finished.

The main gap between plan and execution was pacing. Work was not spread evenly across the sprint. In Sprint 1 the first story only closed around six minutes in, and the remaining four closed within a few minutes of each other near the 10:49 to 10:50 mark. That bunching would be a warning sign in a real team, since it usually means items are being held open and closed in a batch instead of finished one at a time.

4. What insights did the burndown chart give about your team's capacity?

For Sprint 1, the remaining-work line stayed above the guideline for the first half of the sprint and then dropped sharply, finishing at zero around 10:50 against a planned end of 11:00. For Sprint 2, both issues were closed within about a minute of the sprint starting, against a planned end of 11:12. In both cases the team finished well before the deadline, which suggests the committed load was below the team's capacity, especially in Sprint 2 with only 8 points.

The charts also show a limitation in how capacity was measured. They track issue count, not story points, so finishing a 3-point story looks the same as finishing an 8-point one. Switching the chart to story points would give a more honest picture of how much work was burned. Using the observed numbers, the team completed 34 points in Sprint 1 and 8 points in Sprint 2. A reasonable next step is to treat roughly 34 points as the baseline velocity and plan the next sprint closer to that figure instead of committing only 8, while aiming for a more gradual burndown by finishing and reviewing stories one at a time.